Assignment no. 1

Think about your dining experiences. Have you ever wished you could view a menu, check table availability, and place an order all from your phone to avoid long queues? This Restaurant App is the answer.

· **As a Customer:** "I'm hungry and in a hurry. I wish I could browse the daily specials and book a table for 7:00 PM so my food is ready when I arrive."

· **As a Restaurant Manager:** "I have a great menu but struggle with phone-in orders. I'd love to digitize my offerings and manage reservations more efficiently."

Task: Interview yourself as a potential customer. What features would make you download this app? This brainstorming is your "client meeting."

Deliverables

1. **SRS Document**: A detailed document defining every aspect of the full Restaurant system.
2. **UML Diagrams**: Visual designs showing how the system pieces (Diners, Managers, Orders) fit together.
3. **MVP Frontend**: A working React Native app prototype bringing the core screens to life.

Question 1: Software Requirements Specification (SRS)

1. Introduction

1.1. Purpose

This document defines the requirements for "ZEST", a mobile restaurant application. The main purpose of this app is to make it easy for customers to order food for delivery, reserve a table for dine-in, and interact with the restaurant using a modern interface that includes an AI chatbot for food recommendations

1.2. Scope

The following functionalities are in scope for the initial version of the app:

- User account creation and login system.
- Viewing the menu with category filters and a search bar.
- Adding food to a cart and placing delivery orders (with Cash on Delivery or Mobile Wallet options).
- Reserving a table for dine-in and a QR code scanner to claim a table.
- feature to ping the waiter (for water or the bill) or call the delivery rider.
- A profile screen showing reward points and past order history.
- An AI Chatbot feature to give customers smart food pairings.
- A bill-splitting calculator in the cart.
- Feedback Option

1.3. Definitions and Acronyms

- **MVP:** Minimum Viable Product (the first working version of our app).
- **COD:** Cash on Delivery.
- **AI Concierge:** A smart chatbot inside the app that helps users decide what to eat.
- **UI:** User Interface.
- **ZEST Points:** The reward points users earn when they order food.

2. Overall Description

2.1. User Roles

Role	Key Permissions & Goals
Customer	**Can create an account, search the menu, place orders, book tables, use the AI chatbot, split bills, and check their past orders, give feedback.**
Restaurant Manager	**Can see all the incoming orders, manage table bookings, and update the menu items.**
Delivery Rider	**Gets the delivery address and can be contacted by the customer directly through the app if the order is late.**

2.2. User Stories

- **As a Customer**, I want to search for food and filter by categories so I can find my Favorite meals quickly.
- **As a Customer**, I want to book a table for my friends and see a countdown timer so I know my reservation is confirmed.
- **As a Customer**, I want to scan a QR code on my table so the app knows where I am sitting and I can call the waiter to my exact table.
- **As a Customer**, I want to split the bill on the app so I don't have to calculate the math manually with my friends.
- **As a Restaurant Manager**, I want to see digital reservations so I don't have to manage them on phone calls.

4. Functional Requirements

- **FR-1: User Authentication**

- The system must allow users to create an account using their name, email, and password.

- The system must allow users to log in and show an error if the fields are empty.

- **FR-2: Menu and Searching**

- The app must display a list of food items with their names, pictures, and prices.
- The system must let users filter the menu by clicking category buttons (like Pizza, Burger, Pasta).
- The system must have a search bar so users can type what they want and see results instantly.

- **FR-3: Cart and Checkout**

- The system must allow users to add items to their cart and remove them.
- The app must calculate the subtotal, add a delivery fee, and show the final total.
- The app must include a "Split Bill" calculator so users can divide the total cost among their friends.
- The system must give users payment options, specifically Cash on Delivery (COD), EasyPaisa, or JazzCash.

- **FR-4: Dine-In & Table Management**

- The system must allow users to book a table by entering their name, phone number, and party size.
- The app must show a countdown timer when a table is actively reserved.
- The system must allow users to scan a QR code to claim their specific table.
- The system must let seated customers ping the staff for "Water," "Order," or the "Bill."

- **FR-5: AI Concierge**

- The system must include a chat screen where an AI can talk to the user.
- The AI must read the user's message and reply with specific food recommendations based on keywords (like suggesting a burger if they say they are hungry).

- **FR-6: User Profile & History**

- The system must save and display the user's past delivery orders.
- The system must calculate and award "ZEST Points" (loyalty points) based on the order total

4. Non-Functional Requirements

- **NFR-1: Usability (User Experience)**

- The app must be very easy to navigate. A new user should be able to figure out how to order food or book a table within 2 minutes of using it.
- The UI must have a premium "luxury" feel, using a dark theme with gold glowing accents so it looks like a high-end restaurant app.

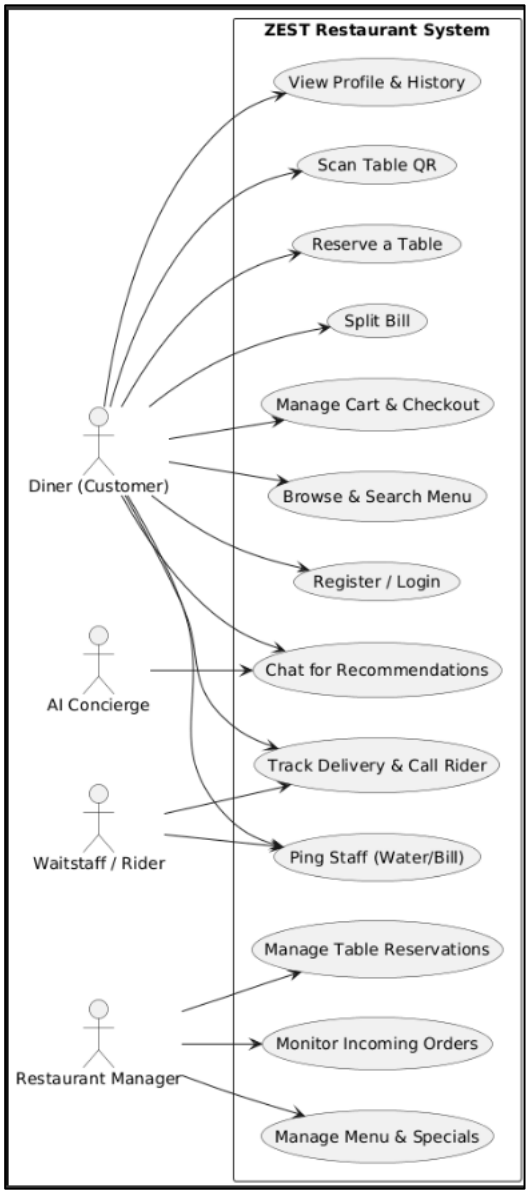
- **NFR-2: Performance**
  - The app should be fast. When a user clicks a category or searches for an item, the menu should update almost instantly without any lag.
  - The AI chatbot should reply within 1.5 to 2 seconds so it feels like a real conversation.
- **NFR-3: Reliability**
  - The app should not crash when the user switches between screens quickly or adds multiple items to the cart.
- **NFR-4: Security**
  - User passwords and email addresses must be kept secure. (For the MVP, we are simulating this, but in the final version, we will use Firebase authentication so we don't store raw passwords).
- **NFR-5: Compatibility**
  - Because it is built using React Native, the app must work smoothly on both Android and iOS smartphones without the layout breaking.

5. Database Schema (NoSQL)

Collection Name	Fields	Description
Users	userId, fullName, email, password, address, zestPoints	Stores the user's account details, delivery address, and reward points.
Menu	itemId, name, category, price, image	Stores the food items we show on the home screen.
Orders	orderId, userId, cartItems, totalPrice, paymentMethod, status	Stores all the delivery orders placed by customers.
Reservations	reservationId, userId, guestName, numberOfPeople, time	Stores the details of table bookings for dine-in customers.

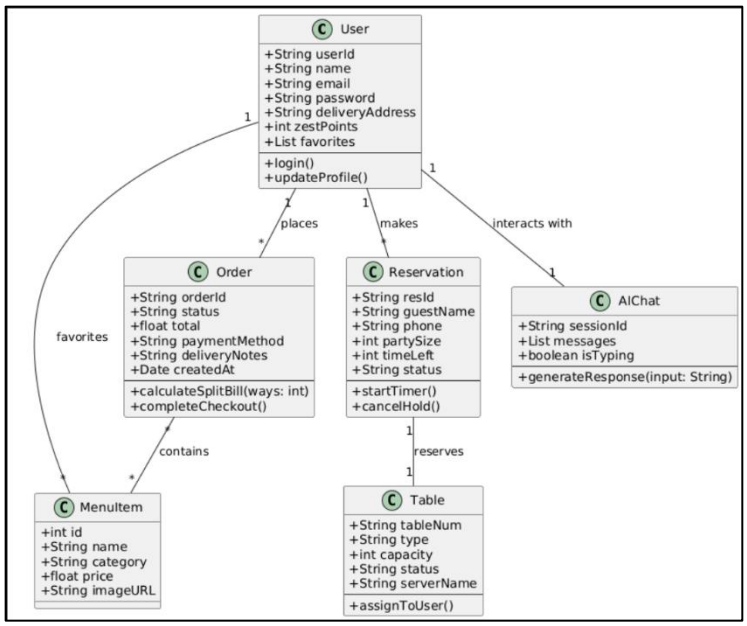
6. UML Diagrams (Placeholders)

USECASE Diagram:



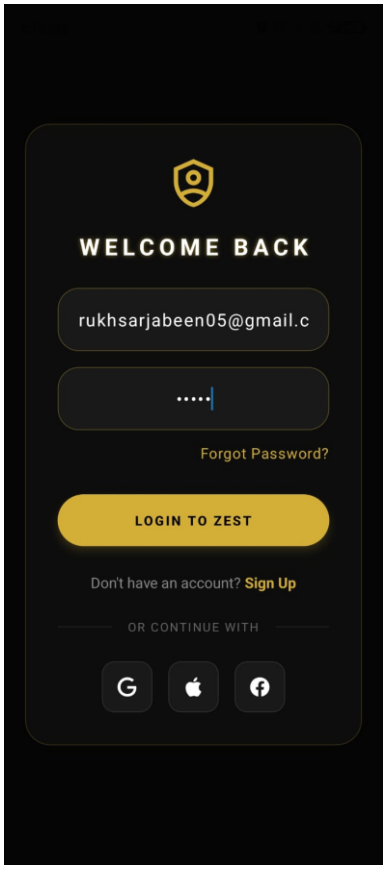
2. Class Diagram (Entity Relationship)

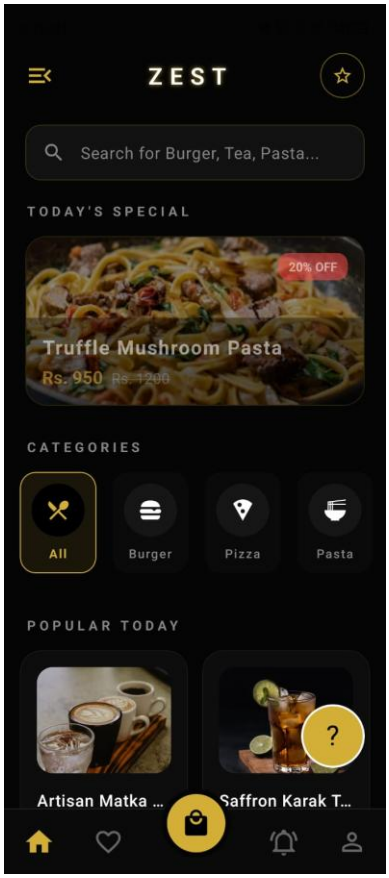
This diagram maps out the internal data structures and how they relate to one another based on the states we built in the app.

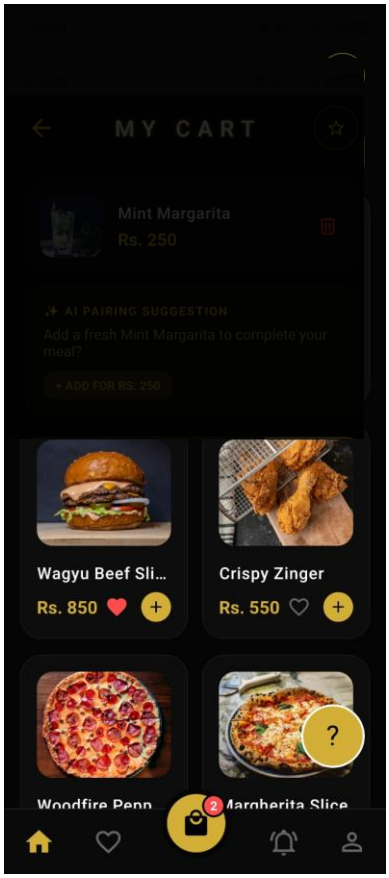


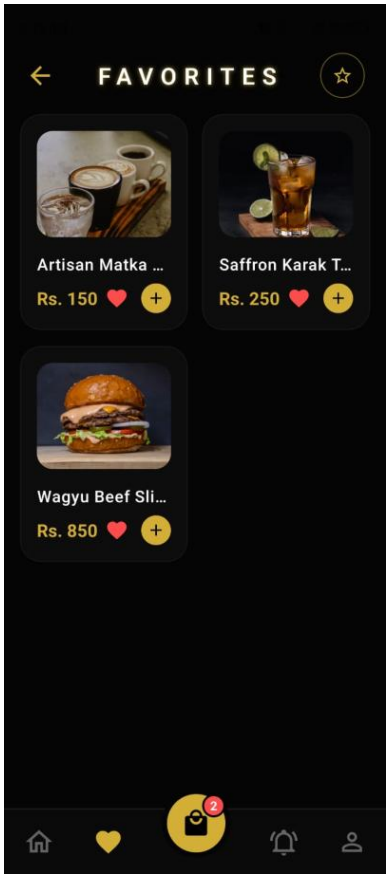
7. MVP Frontend Development (React Native)

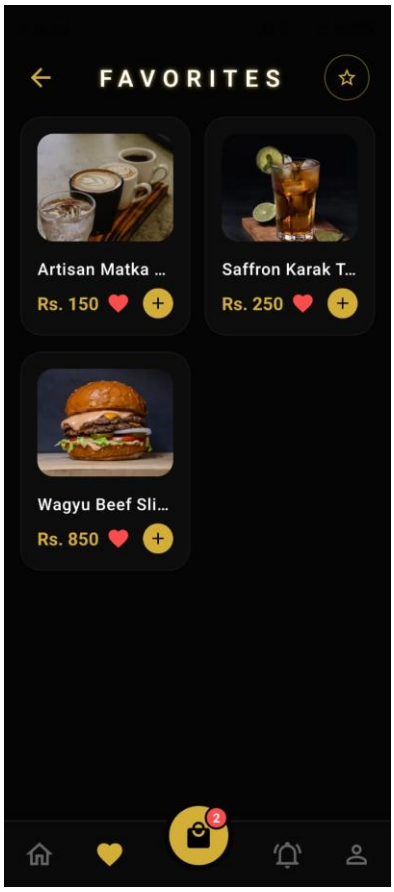
7.1. Login Screen

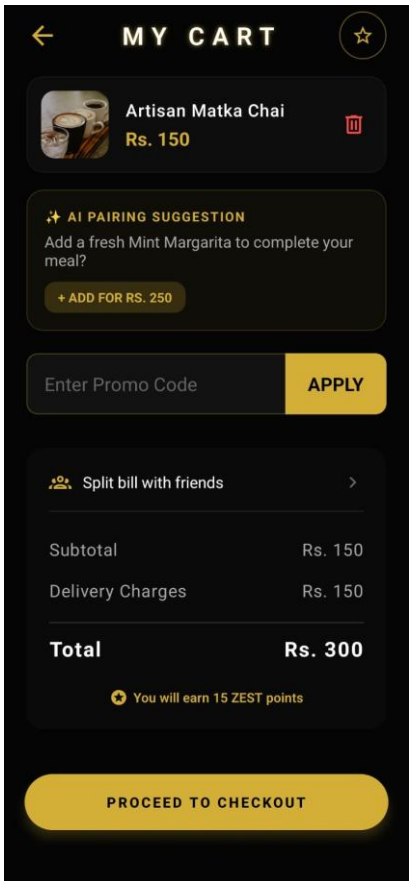


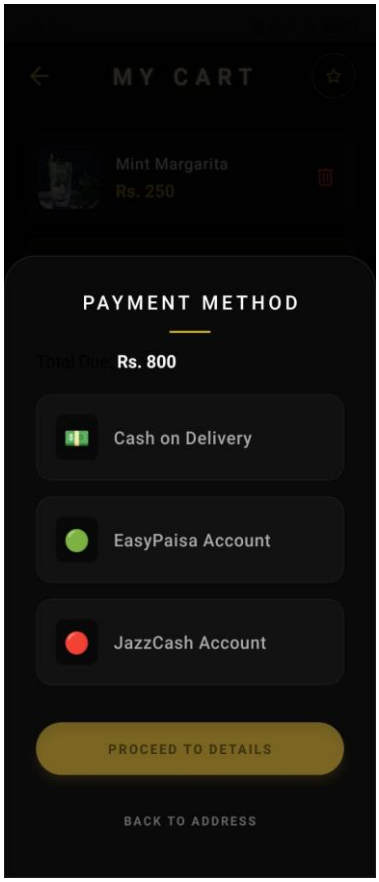

ZEST
Search for Burger, Tea, Pasta...
TODAY'S SPECIAL
20% OFF
Truffle Mushroom Pasta
Rs. 950 Rs. 1200
CATEGORIES
All
Burger
Pizza
Pasta
POPULAR TODAY
Artisan Matka ...
Saffron Karak T...

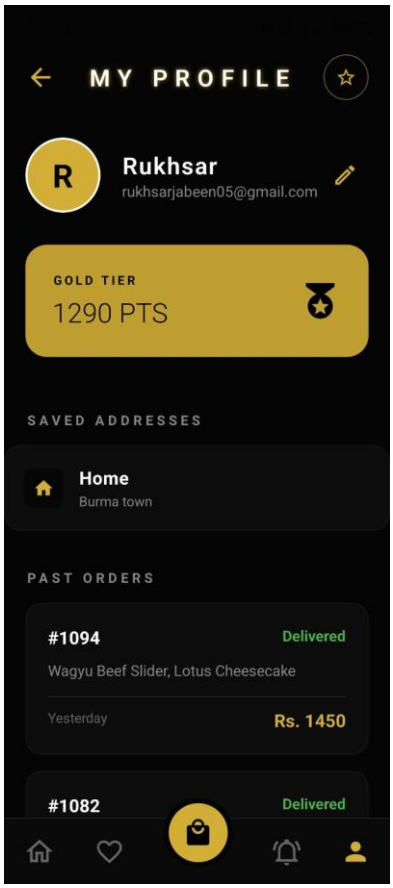

MY CART
Mint Margarita
Rs. 250
AI PAIRING SUGGESTION
Add a fresh Mint Margarita to complete your meal?
+ ADD FOR RS. 250
Wagyu Beef Sli...
Rs. 850
Crispy Zinger
Rs. 550
Woodfire Penn
Margherita Slice


FAVORITES
Artisan Matka ...
Rs. 150
Saffron Karak T...
Rs. 250
Wagyu Beef Sli...
Rs. 850

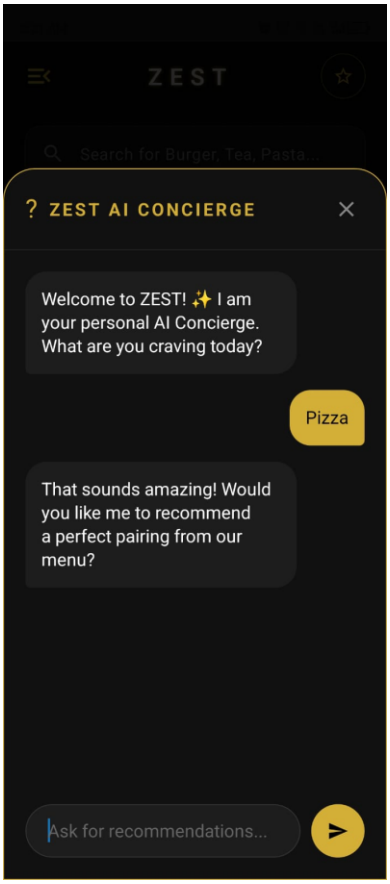

FAVORITES
Artisan Matka ...
Rs. 150
Saffron Karak T...
Rs. 250
Wagyu Beef Sli...
Rs. 850

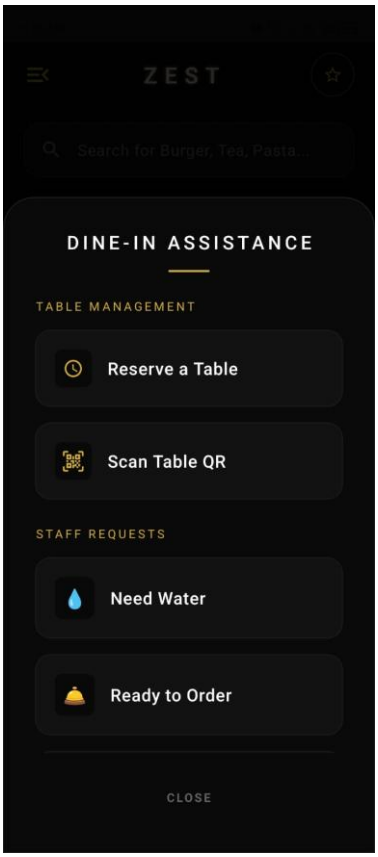

MY CART
Artisan Matka Chai
Rs. 150
AI PAIRING SUGGESTION
Add a fresh Mint Margarita to complete your meal?
+ ADD FOR RS. 250
Enter Promo Code
APPLY
Split bill with friends
Subtotal Rs. 150
Delivery Charges Rs. 150
Total Rs. 300
You will earn 15 ZEST points
PROCEED TO CHECKOUT

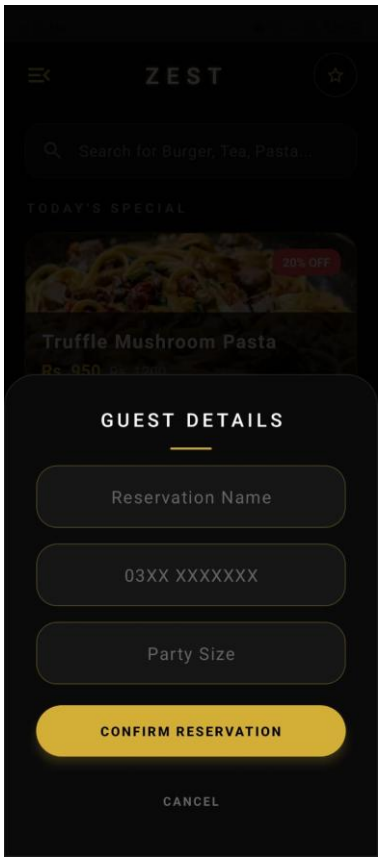

MY CART
Mint Margarita
Rs. 250
PAYMENT METHOD
Total Due Rs. 800
Cash on Delivery
EasyPaisa Account
JazzCash Account
PROCEED TO DETAILS
BACK TO ADDRESS

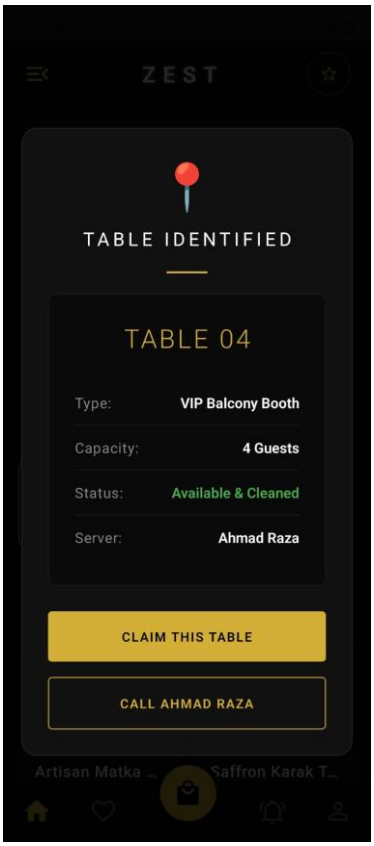

MY PROFILE
R
Rukhsar
rukhsarjabeen05@gmail.com
GOLD TIER
1290 PTS
SAVED ADDRESSES
Home
Burma town
PAST ORDERS
#1094
Delivered
Wagyu Beef Slider, Lotus Cheesecake
Yesterday
Rs. 1450
#1082
Delivered

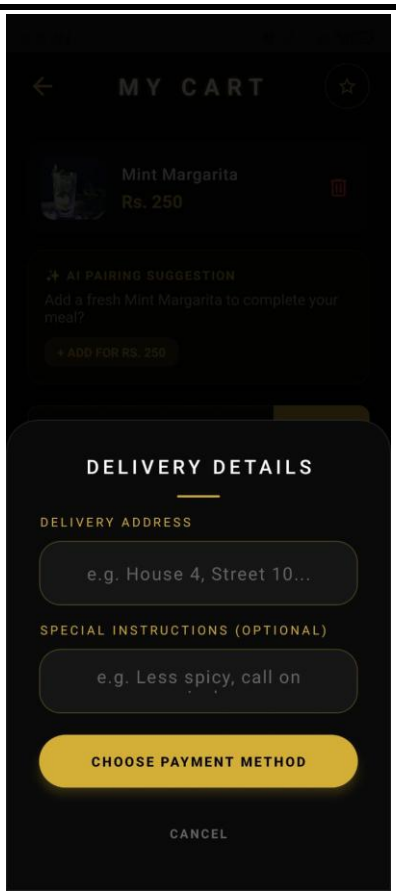
GUEST EXPERIENCE
Tap to rate your visit to ZEST
I love the taste of burgers
SEND FEEDBACK
CLOSE

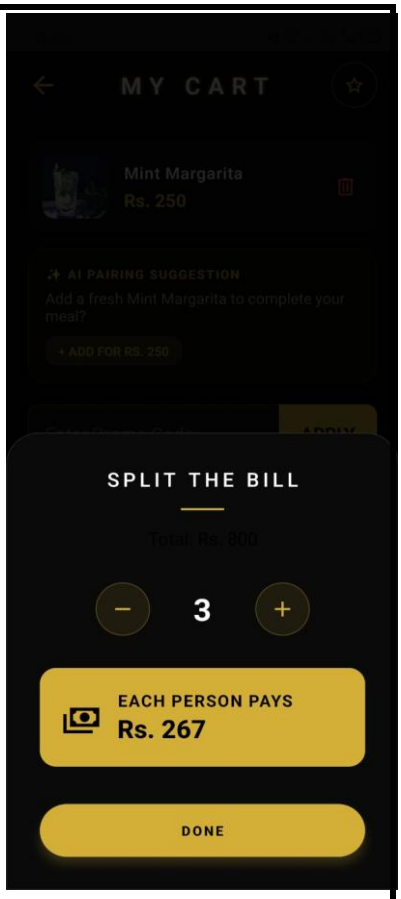

ZEST
ZEST AI CONCIERGE
Welcome to ZEST! ✨ I am your personal AI Concierge. What are you craving today?
Pizza
That sounds amazing! Would you like me to recommend a perfect pairing from our menu?
Ask for recommendations...


ZEST
DINE-IN ASSISTANCE
TABLE MANAGEMENT
Reserve a Table
Scan Table QR
STAFF REQUESTS
Need Water
Ready to Order
CLOSE


ZEST
TODAY'S SPECIAL
20% OFF
Truffle Mushroom Pasta
GUEST DETAILS
Reservation Name
03XX XXXXXXX
Party Size
CONFIRM RESERVATION
CANCEL


ZEST
TABLE IDENTIFIED
TABLE 04
Type: VIP Balcony Booth
Capacity: 4 Guests
Status: Available & Cleaned
Server: Ahmad Raza
CLAIM THIS TABLE
CALL AHMAD RAZA





Submission Requirements

- Create a GitHub repository named restaurant-app-mvp.
- Place the **SRS** and **UML** in a folder named A1.
- Push your code to the main branch.
- Include a README.md with installation steps.